

A. **Primes.** An integer is **prime** if it is greater than 1 and its only divisors are:

In particular, 1 is not prime.

which we call the:

For example, the primes less than 50 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

An integer greater than 1 that is not prime is called **composite**.

Compositeness can be detected by a **nontrivial** factorization.

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Theorem. Let p be prime and a an integer. Then $p \perp a$ iff:

Proof. The only positive divisors of p are 1 and p .

Thus the greatest common divisor of p and a is one of the following.

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□

Theorem. Let p be prime and $a_1, a_2, \dots, a_n$ integers. If $p \mid a_1 a_2 \dots a_n$, then:

Proof. We argue by induction on the number n of integers in the product.

Base Case. If $n = 1$ then the theorem states:

which is obviously true.

Inductive Hypothesis. Let positive integer n be given.

Assume the theorem holds for products of n integers.

Inductive Step. We need to prove the theorem holds for:

Let $a_1, a_2, \dots, a_n, a_{n+1}$ be integers such that:

If $p \mid a_{n+1}$ then we are done. So consider the case that:

Because p is prime this implies:

Since further:

we conclude by Euclid's lemma that:

By the inductive hypothesis:

□

Example 1. Prove the only prime p for which $p + 4$ is a perfect square is $p =$

Solution. Suppose that p is prime $p + 4$ is a perfect square. Then:

$$p + 4 =$$

And thus rewriting:

$$p =$$

This will be a nontrivial factorization except if

$$n - 2 =$$

But in this case

Example 2. Let p be prime. Prove that $\sqrt{p}$ is **irrational**.

Rational means expressible as a fraction of integers.

Solution. This is a famous argument by contradiction.

Suppose towards a contradiction that

$$\sqrt{p} =$$

where a, b are integers and common factors have been cancelled so:

Then

$$p =$$

Thus $p \mid$

Because p is prime, this implies:

So we write:

And then

$$pb^2 =$$

Thus $p \mid$

Because p is prime, this implies:

This contradicts our assumption that:

because we have shown:

□

B. Fundamental Theorem of Arithmetic. Primes are the integer building blocks.

Any integer can be written as a product of powers of primes. For example

$$360 =$$

This is called a **prime decomposition**.

Fundamental Theorem of Arithmetic. Let n be an integer greater than 1.

Then n has a prime decomposition

$$n =$$

where:

Further, the primes and their powers in this decomposition are:

Proof. We use strong induction on n to prove the theorem.

Base case. If $n = 2$ then the prime decomposition is:

and is evidently unique.

Strong Inductive Hypothesis. Let integer n greater than 2 be given.

Assume all integers in $[2 \dots n - 1]$ have unique prime decompositions.

Inductive Step. We need to prove n has a unique prime decomposition.

If n is prime, $n = p$, then:

If n is composite, then n has a nontrivial factorization

$$n =$$

By the strong inductive hypothesis, both a and b have prime decompositions.

Thus $n = ab$ can also be written as a:

and hence n has a prime decomposition.

To prove the decomposition is unique, suppose we have two decompositions.

$$n =$$

Since p_1 is prime and divides $n = q_1^{l_1} q_2^{l_2} \cdots q_s^{l_s}$ it follows that:

Because q_i is prime, and p_1 is positive and not 1, this implies:

Reorder so:

Divide both decompositions by $p_1 = q_1$ to obtain:

$$n/p_1 =$$

By the strong inductive hypothesis, n/p_1 has a:

So, after reordering if necessary, we have:

Then after reordering if necessary, the two prime decompositions of n also match.

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